

Chapter 5

Combinational Logic

组合逻辑

Logic circuit

Combinational Logic circuit

(组合逻辑电路：任意时刻的输出仅取决于该时刻的输入，与电路原来的状态无关)

Sequential Logic circuit

(时序逻辑电路：任意时刻的输出取决于该时刻的输入信号以及电路原来的状态)

5.1 Basic Combinational Circuits

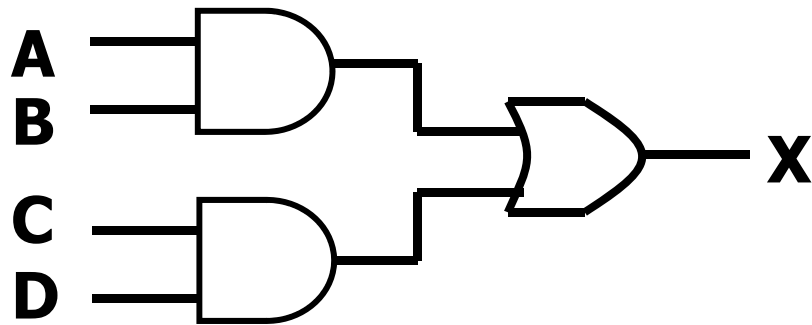
- AND-OR（与-或电路/SOP电路）
- AND-OR-Invert（与-或-非电路/POS电路）
- XOR（异或电路）
- XNOR（同或电路）
- NAND（与非门）
- NOR（或非门）

1 AND-OR Logic

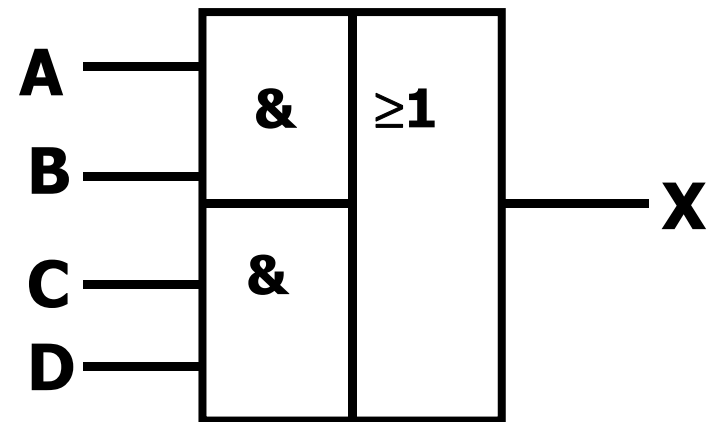
$$X=AB+CD$$

An AND-OR circuit directly implements SOP expression, assuming the complements of the variables are available.

与或电路直接实现了SOP的布尔表达式



ANSI standard distinctive shape symbols



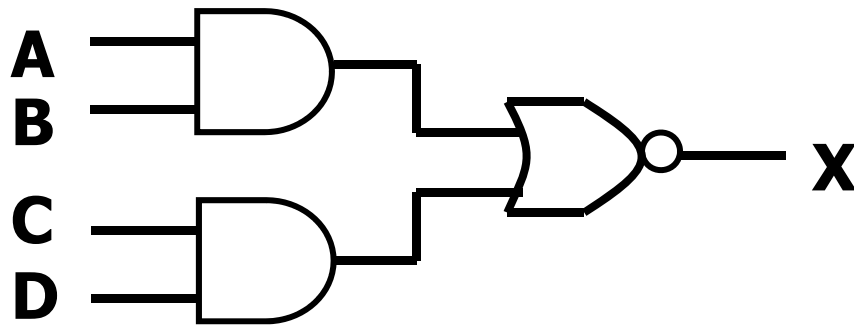
ANSI standard rectangular outline symbols

2 AND-OR-Invert Logic

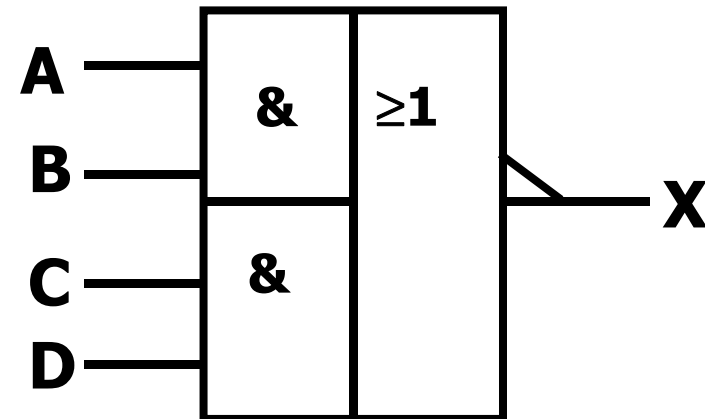
$$X = \overline{AB + CD} = (\overline{A} + \overline{B})(\overline{C} + \overline{D})$$

An AND-OR-Invert can be used to implement **POS** expression.

与或非电路直接实现了POS形式的布尔表达式



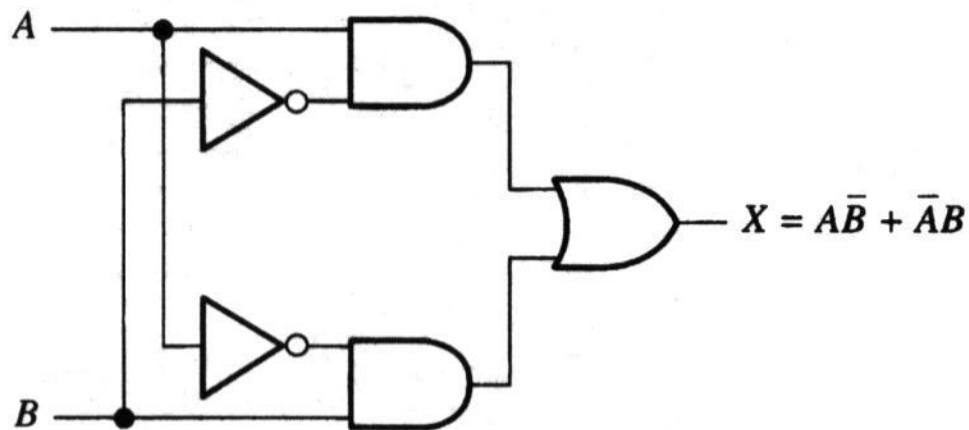
ANSI standard distinctive
shape symbols



ANSI standard rectangular
outline symbols

3 Exclusive-OR (XOR)异或

$$X = \bar{A}B + A\bar{B} = A \oplus B$$

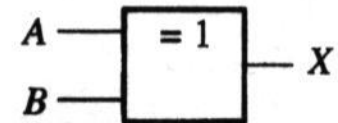


(a) Logic diagram

Inputs		Output
A	B	X
0	0	0
0	1	1
1	0	1
1	1	0



(b) ANSI distinctive shape symbol



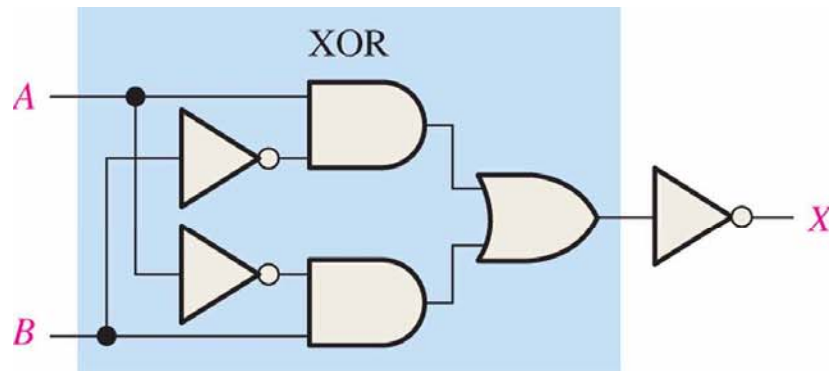
(c) ANSI rectangular outline symbol

- This circuit is considered a type of logic gate with its own unique symbol, it is actually a combination of two AND gates, one OR gate, and two inverters.

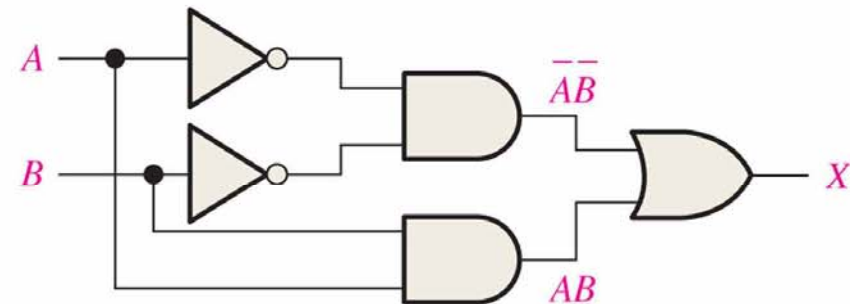
4 Exclusive-NOR (XNOR)同或

$$X = AB + \bar{A}\bar{B} = \overline{A \oplus B}$$

Inputs		Output
A	B	X
0	0	1
0	1	0
1	0	0
1	1	1



(a) $X = \bar{A}\bar{B} + \bar{A}B$

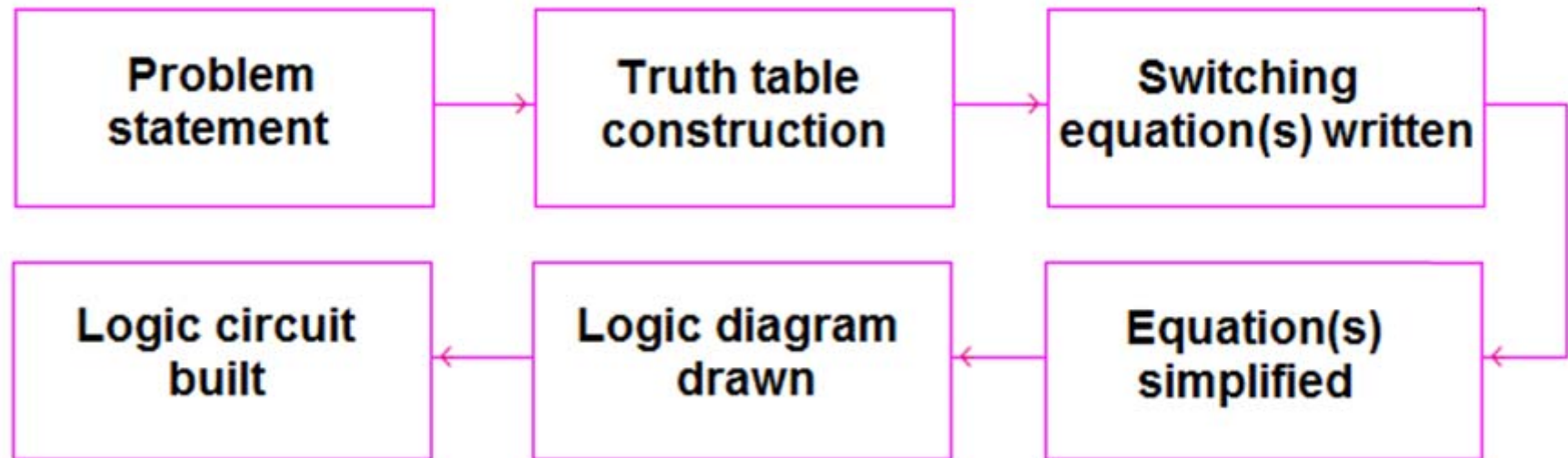


(b) $X = \bar{A}\bar{B} + AB$

5-2 IMPLEMENTING COMBINATIONAL LOGIC

组合逻辑实现

General logic design sequence



(Digital Logic Applications and Design by John M. Yarbrough)

Example : Design a combinational logic so that an output is generated indicating when a majority of three inputs is true.

Solution

(1) Assign input and output variables. The circuit has three inputs and a single output. Let A , B , and C represent the input variables, and X be the output variable.

(2) Construct a truth table. A majority occurs any time two or more of the input variables are true.

Truth table for the combinational logic

Inputs			Output
<i>A</i>	<i>B</i>	<i>C</i>	<i>X</i>
0	0	0	0
0	0	1	0
0	1	0	0
0	1	1	1
1	0	0	0
1	0	1	1
1	1	0	1
1	1	1	1

(3) Derive the logic expression.

$$X = \overline{A}BC + A\overline{B}C + AB\overline{C} + ABC$$

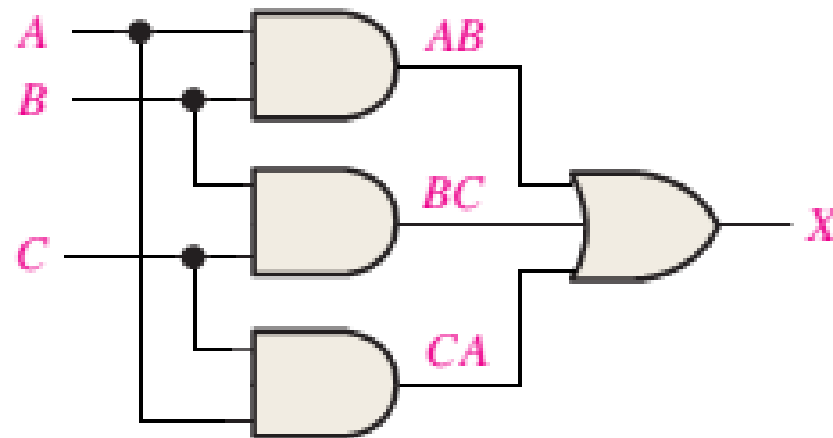
(4) Simplify the logic expression.

$$X = \overline{A}BC + A\overline{B}C + AB\overline{C} + ABC$$

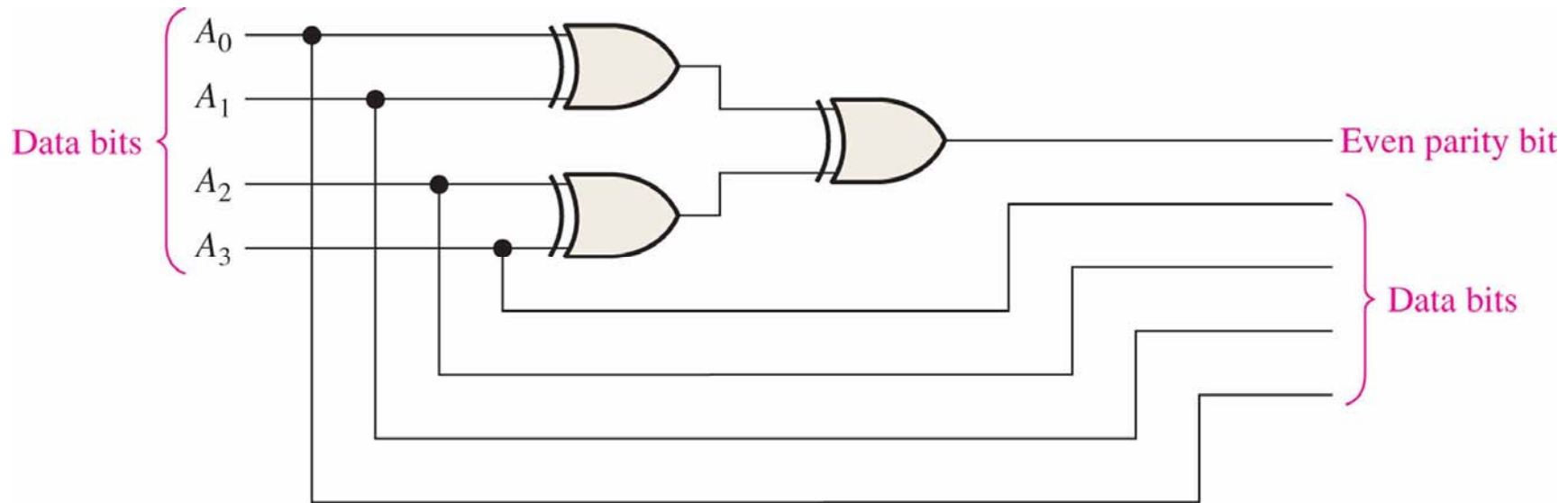
$$= (\overline{A}BC + ABC) + (A\overline{B}C + ABC) + (AB\overline{C} + ABC)$$

$$= BC + AC + AB = AB + BC + CA$$

(5) Draw the logic diagram.



Example: Even-parity check generator



Question: how to design a even-parity checker?

5-3 THE UNIVERSAL PROPERTY OF NAND AND NOR GATES

与非门和或非门的通用特征

The NAND Gate as a Universal Logic Element

- The NAND gate can be used to produce the NOT, AND, OR, and NOR operations. (与非门可以用来构建非门、与门、或门和或非门)

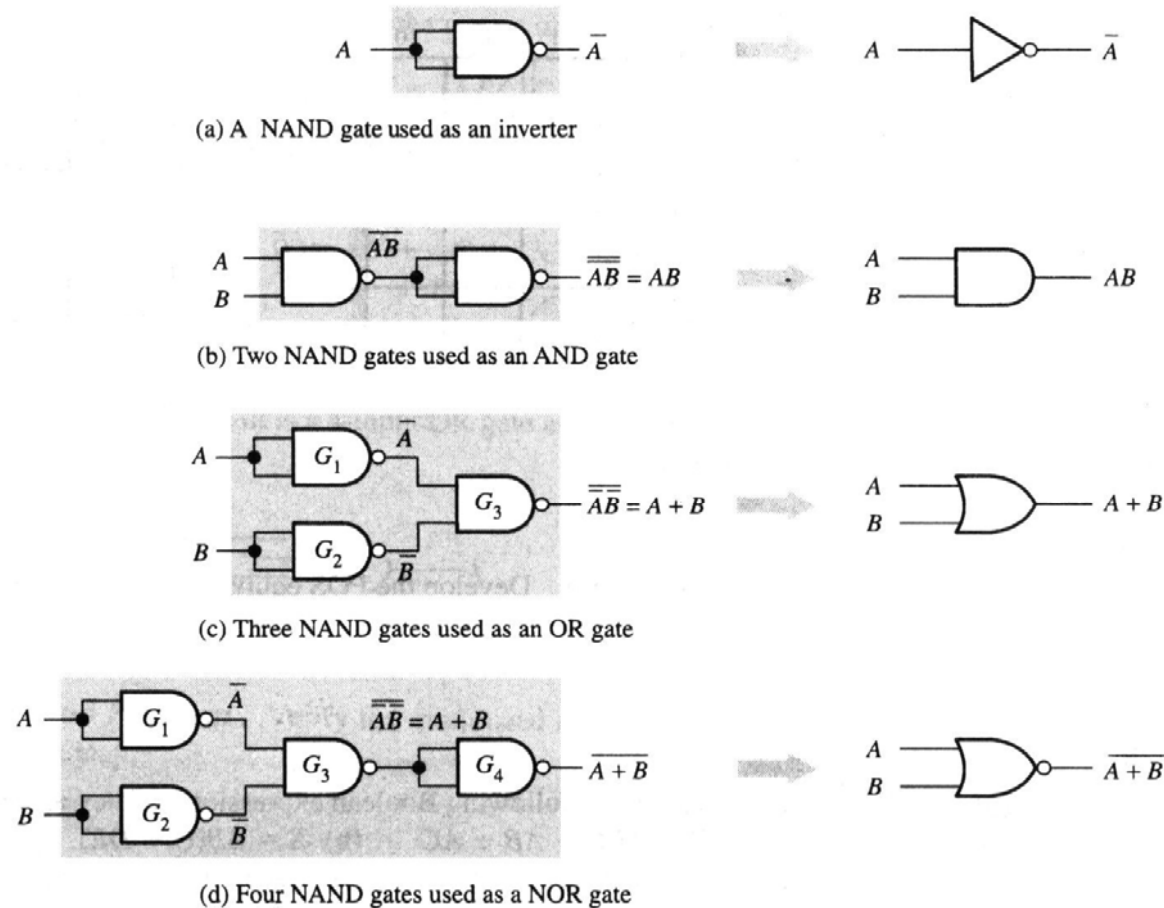


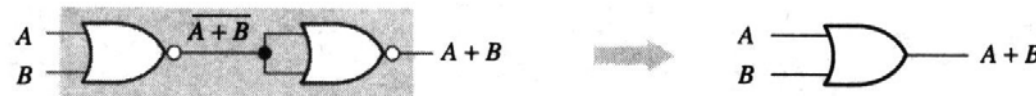
FIGURE 5-18
Universal application of NAND gates.

The NOR Gate as a Universal Logic Element

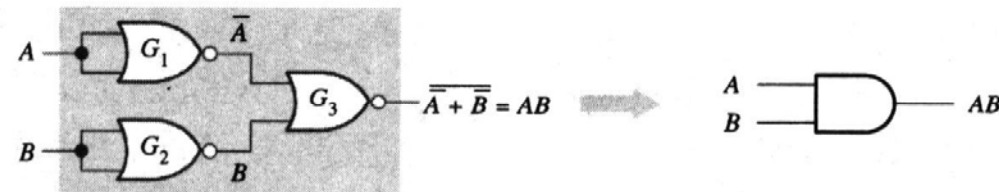
- The NOR gate can be used to produce the NOT, AND, OR, and NAND operations. (与非门可以用来构建非门、与门、或门和与非门)



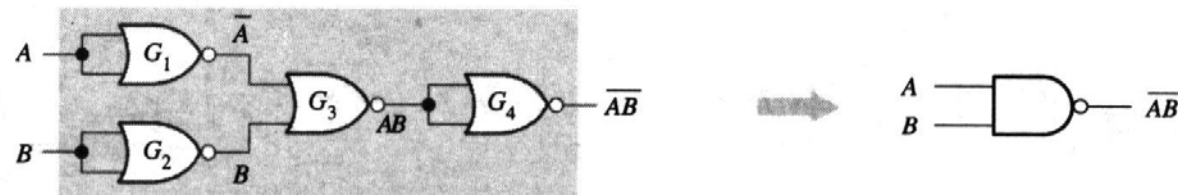
(a) A NOR gate used as an inverter



(b) Two NOR gates used as an OR gate



(c) Three NOR gates used as an AND gate



(d) Four NOR gates used as a NAND gate

FIGURE 5-19

5-4 COMBINATIONAL LOGIC USING NAND AND NOR GATES

使用与非门和或非门的组合
逻辑

5 Combinational Logic Using NAND and NOR Gates

- NAND

equivalent

$$\overline{AB} = \overline{A} + \overline{B} \longrightarrow \text{Negative-OR}$$

- NOR

equivalent

$$\overline{A + B} = \overline{A} \overline{B} \longrightarrow \text{Negative-AND}$$

The **NAND** symbol and the **negative-OR** symbol are called dual symbols. (与非门与非或门是一对对偶电路)

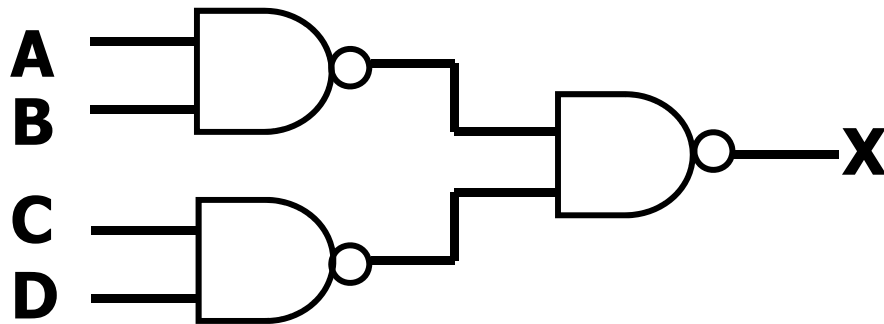
The **NOR** symbol and the **negative-AND** symbol are called dual symbols. (或非门与非与门是一对对偶电路)

Example: Implementing the following expression using NAND gates:

$$X = AB + CD$$

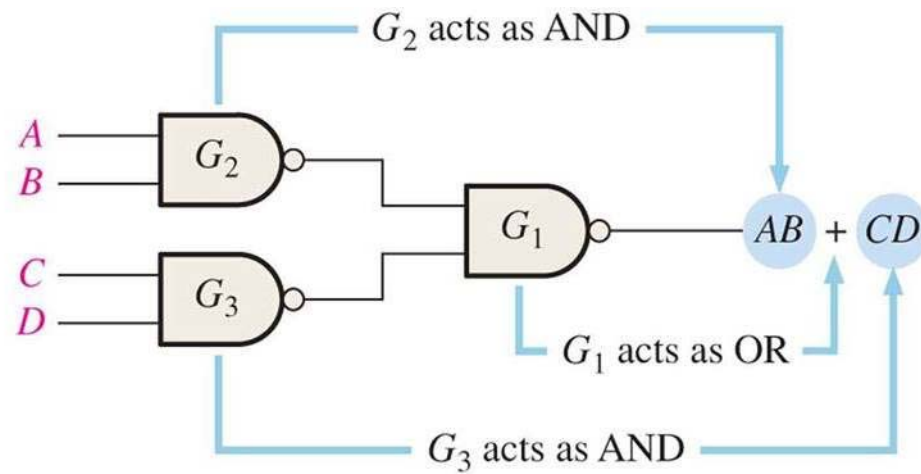
Solution: $X = AB + CD$

$$= \overline{\overline{AB + CD}} = \overline{\overline{AB} \cdot \overline{CD}}$$

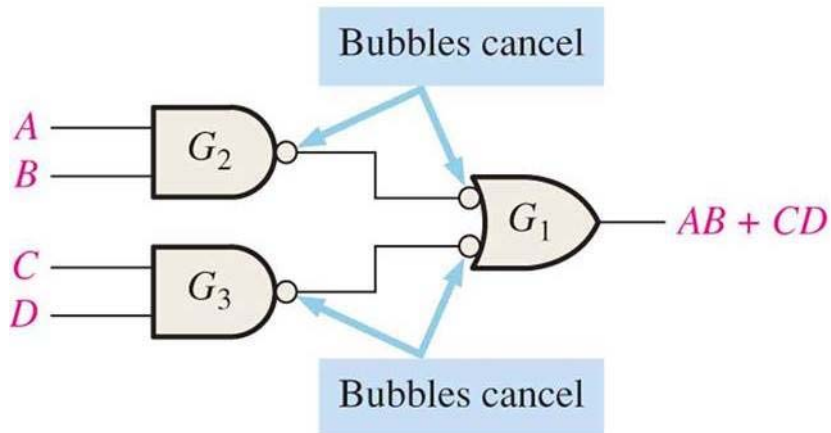


NAND Logic Diagrams Using Dual Symbols

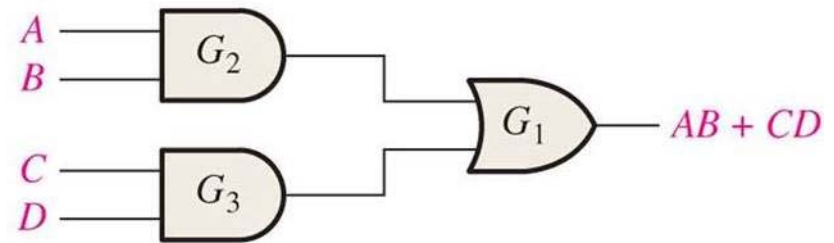
- All logic diagrams using NAND gates should be drawn with each gate represented by either a **NAND symbol** or the equivalent **negative-OR symbol** to represent the operations.
- Use the gate symbols in such a way that **every connection between a gate output and a gate input is either bubble-to-bubble or nonbubble-to-nonbubble.**



(a) Original NAND logic diagram showing effective gate operation relative to the output expression

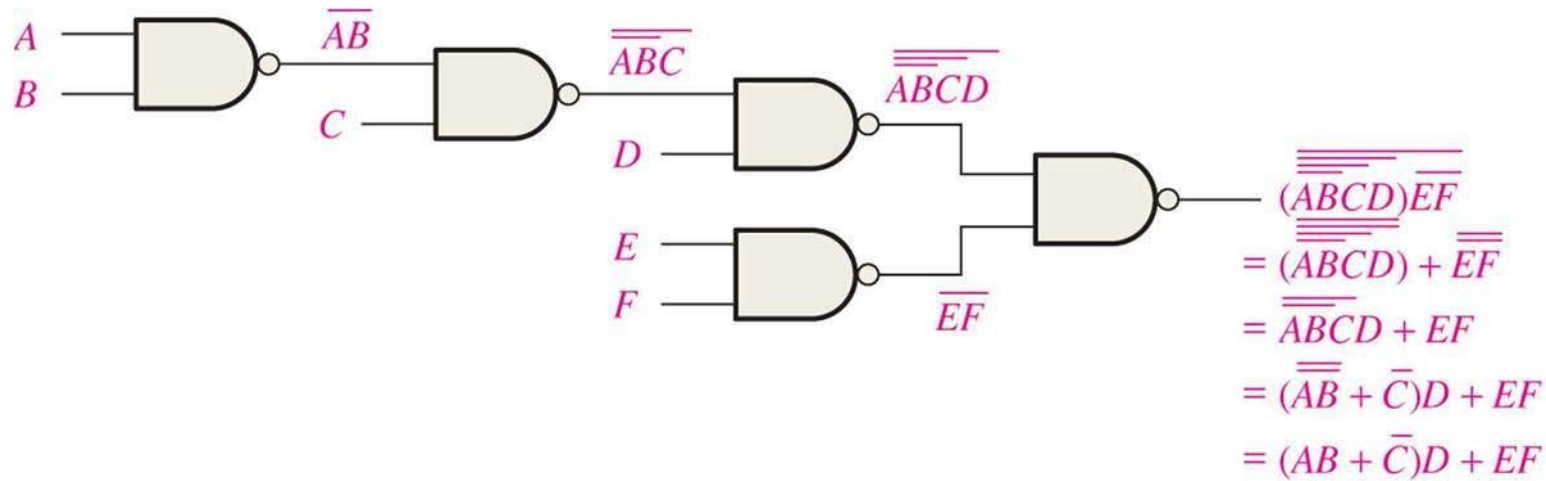


(b) Equivalent NAND/Negative-OR logic diagram

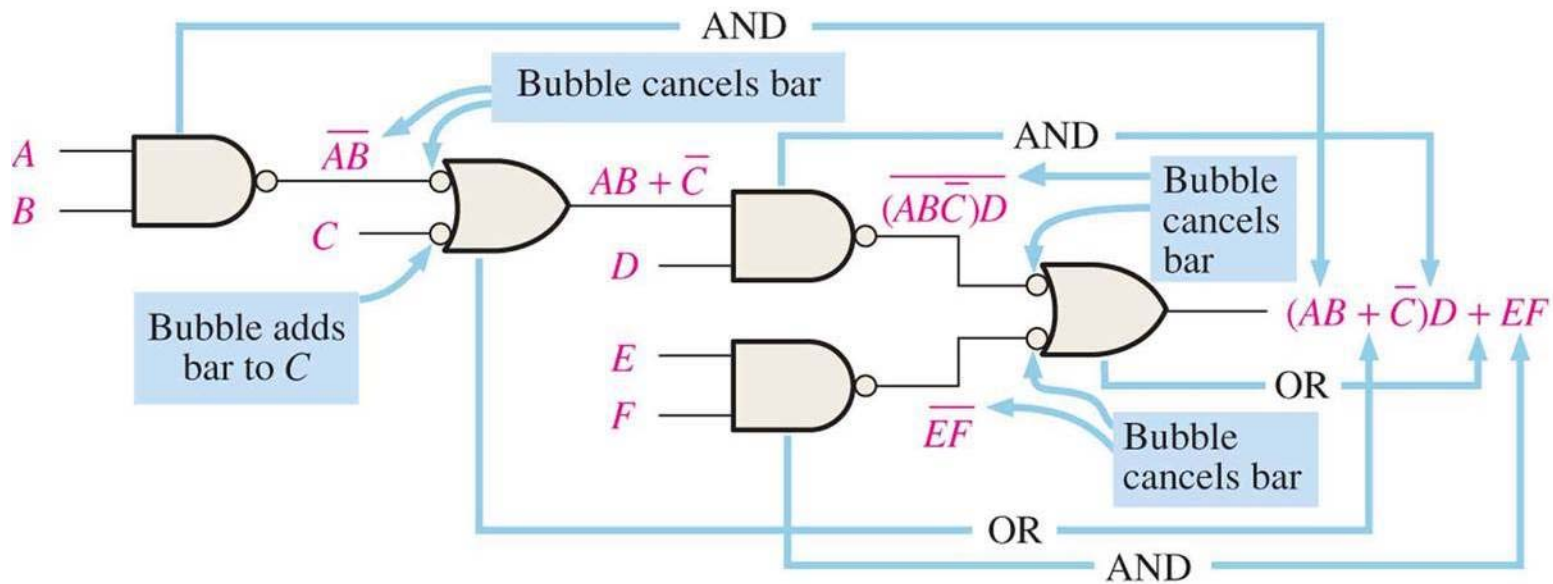


(c) AND-OR equivalent

NAND Logic



(a) Several Boolean steps are required to arrive at final output expression.



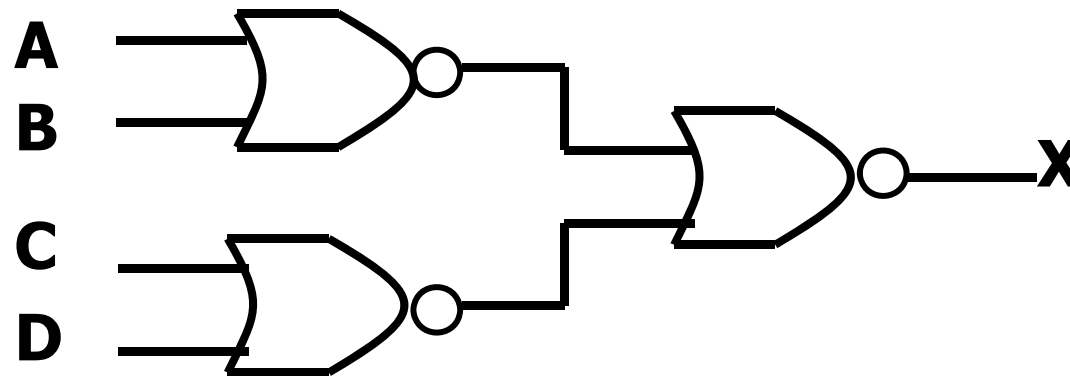
(b) Output expression can be obtained directly from the function of each gate symbol in the diagram.

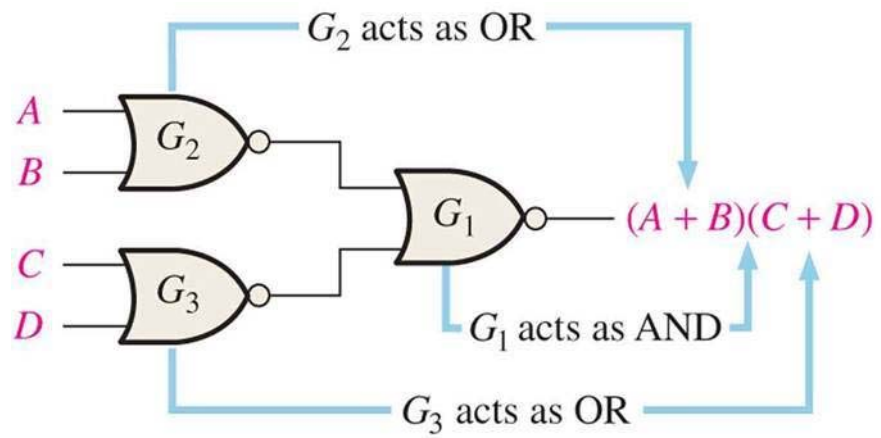
Example: Implementing the following expression using NOR gates:

$$X = (A + B)(C + D)$$

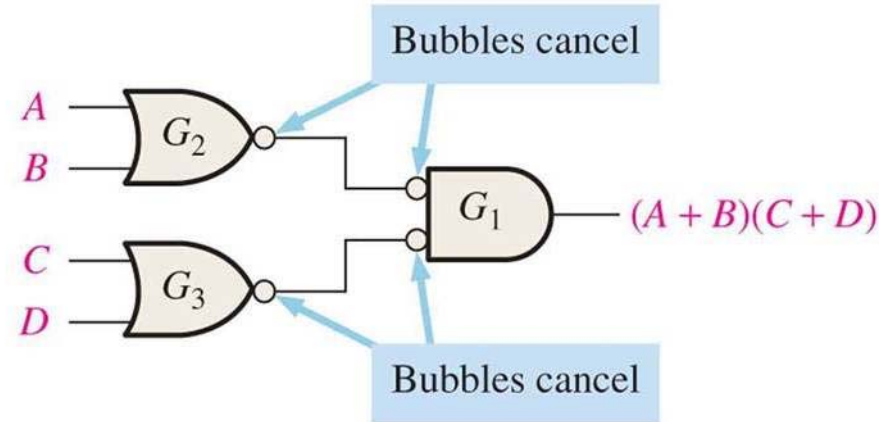
Solution: $X = (A + B)(C + D)$

$$= \overline{\overline{(A + B)(C + D)}} = \overline{A + B + C + D}$$



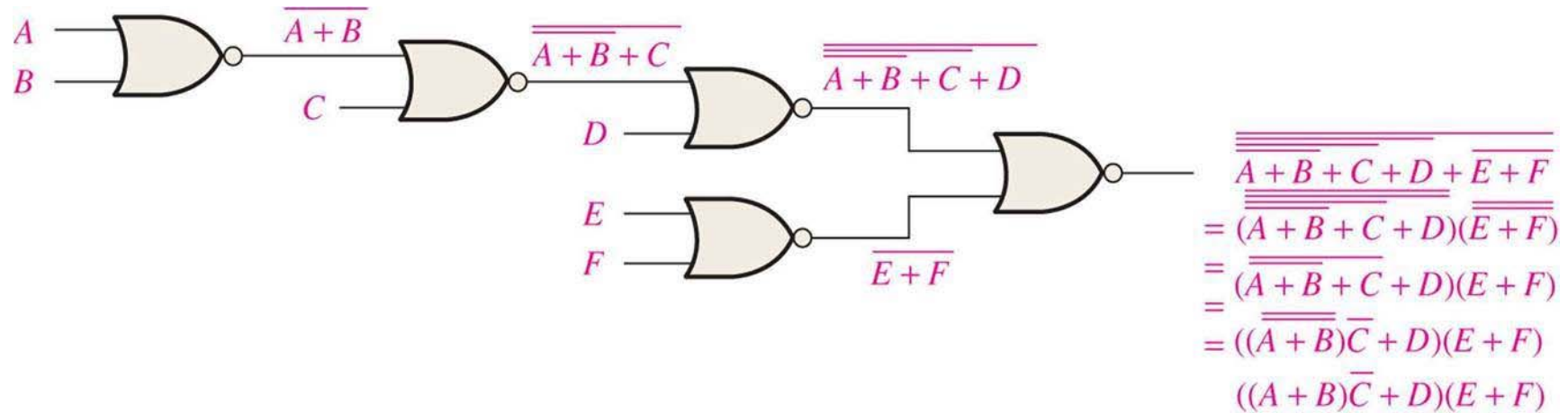


(a)

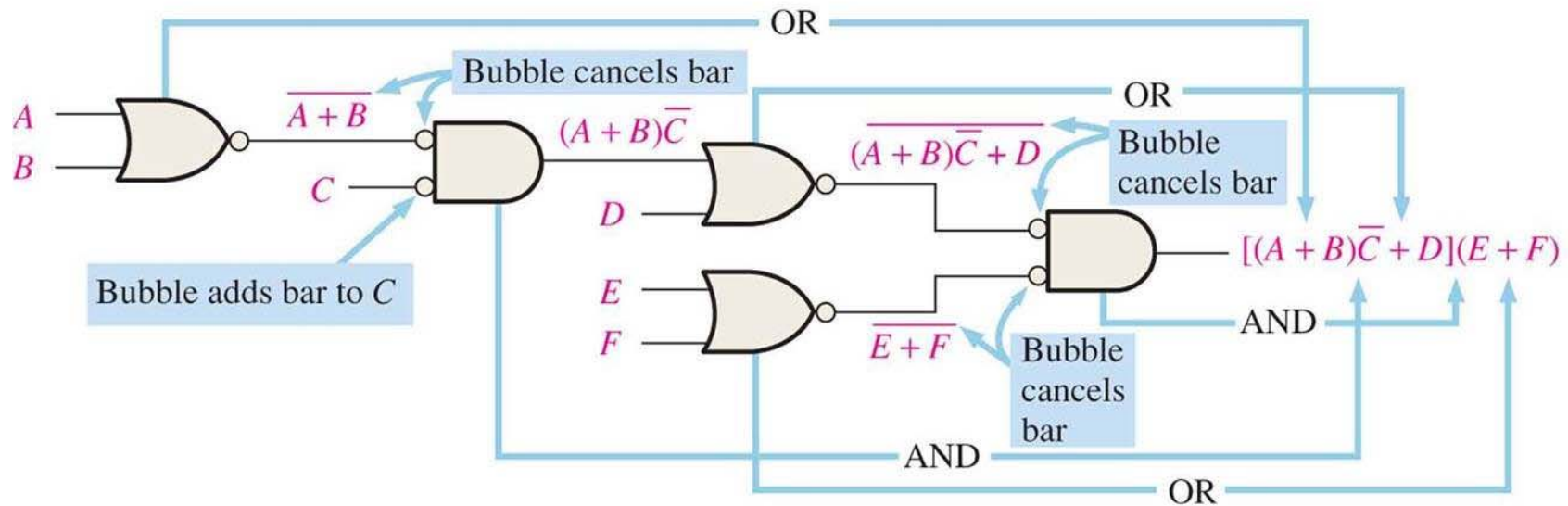


(b)

NOR Logic



(a) Final output expression is obtained after several Boolean steps.

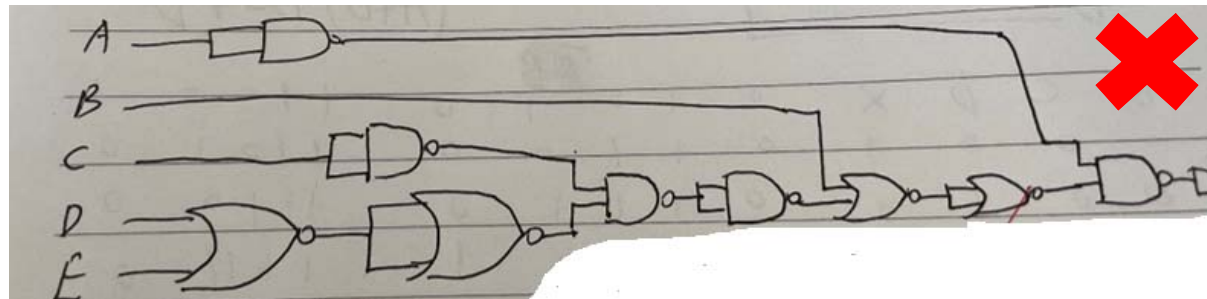
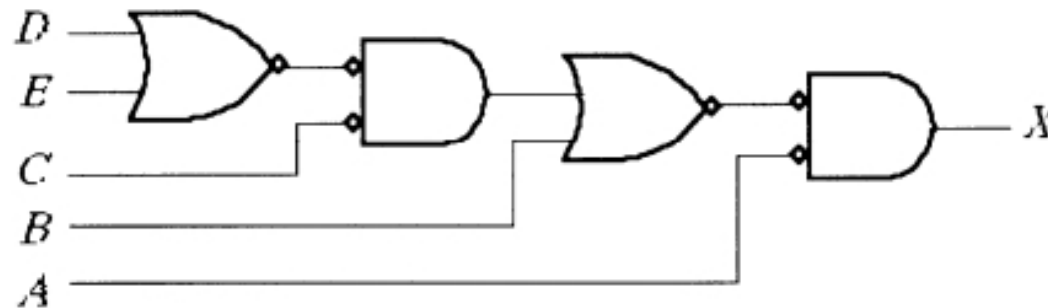


(b) Output expression can be obtained directly from the function of each gate symbol in the diagram.

Example

Implementing the following expression using NOR gates:

$$X = \overline{A}[B + \overline{C}(D + E)]$$



Quiz

1. Assume an SOP expression is $AB + CD$. The equivalent POS expression is

- a. $(A + B)(C + D)$
- b. $(\bar{A} + B)(\bar{C} + D)$
- c. $(\bar{A} + \bar{B})(\bar{C} + \bar{D})$
- d. none of the above

Quiz

2. The truth table shown is for

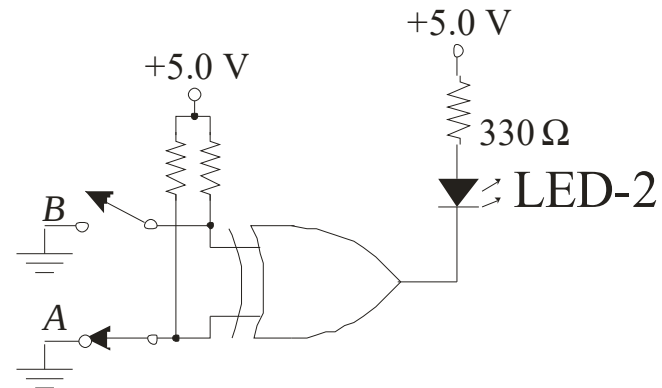
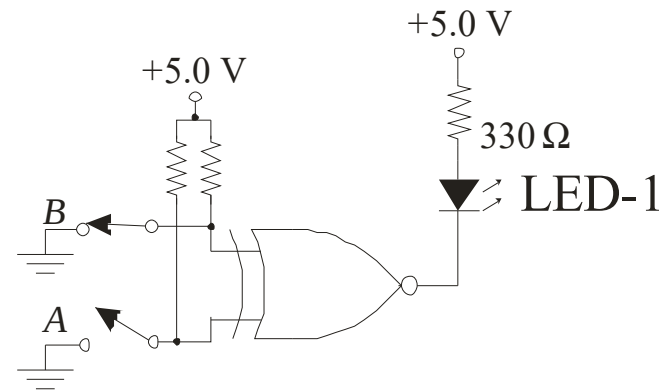
- a. a NAND gate
- b. a NOR gate
- c. an exclusive-OR gate
- d. an exclusive-NOR gate

Inputs		Output
A	B	X
0	0	1
0	1	0
1	0	0
1	1	1

Quiz

3. An LED that should be ON is

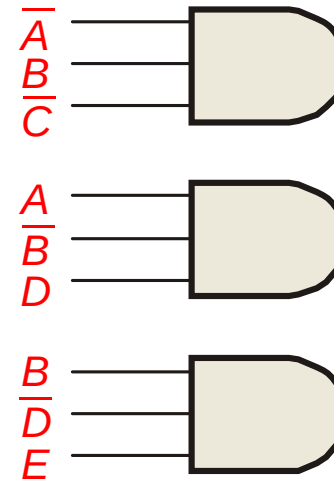
- a. LED-1
- b. LED-2
- c. neither
- d. both



Quiz

4. To implement the SOP expression $X = \bar{A}\bar{B}\bar{C} + A\bar{B}D + B\bar{D}E$, the type of gate that is needed is a

- a. 3-input AND gate
- b. 3-input NAND gate
- c. 3-input OR gate
- d. 3-input NOR gate



Quiz

5. Reading the Karnaugh map, the logic expression is

a. $A\bar{C} + \bar{A}\bar{B}$

b. $\bar{A}B + A\bar{C}$

c. $\bar{A}B + B\bar{C}$

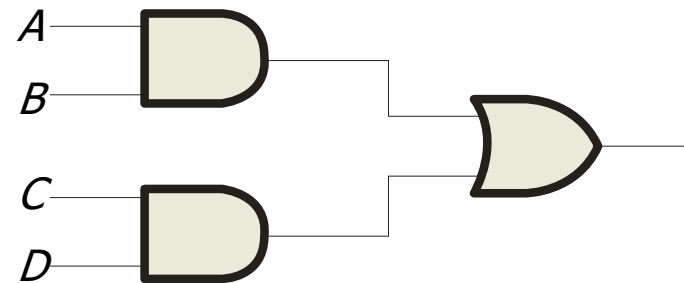
d. $\bar{A}B + \bar{A}\bar{C}$

	$\bar{C}$	C
$\bar{A}\bar{B}$	1	
$\bar{A}B$	1	1
AB		
$A\bar{B}$		

Quiz

6. The circuit shown will have identical logic out if all gates are changed to

- a. AND gates
- b. OR gates
- c. NAND gates
- d. NOR gates



Quiz

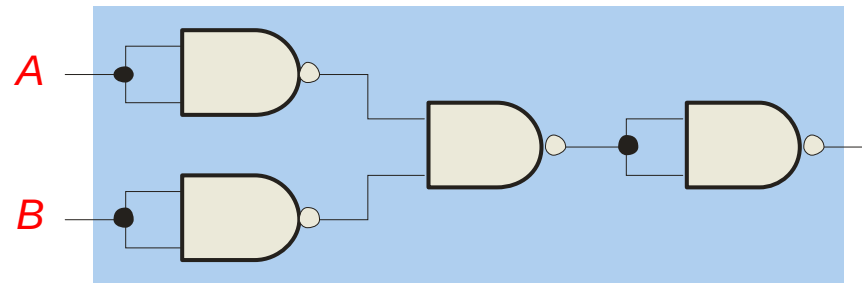
7. The two types of gates which are called *universal gates* are

- a. AND/OR
- b. NAND/NOR
- c. AND/NAND
- d. OR/NOR

Quiz

8. The circuit shown is equivalent to an

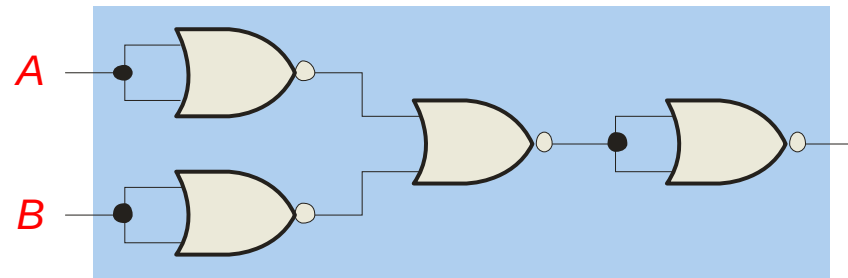
- a. AND gate
- b. XOR gate
- c. NOR gate
- d. none of the above



Quiz

9. The circuit shown is equivalent to

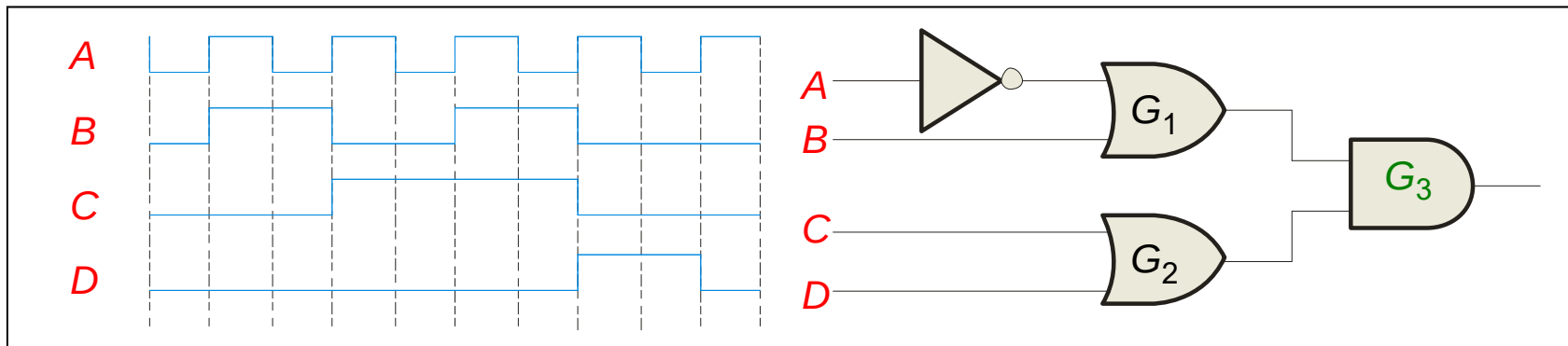
- a. an NAND gate
- b. an XOR gate
- c. an OR gate
- d. none of the above



Quiz

10. During the first *three* intervals for the pulsed circuit shown, the output of

- a. G_1 is LOW and G_2 is LOW
- b. G_1 is LOW and G_2 is HIGH
- c. G_1 is HIGH and G_2 is LOW
- d. G_1 is HIGH and G_2 is HIGH



Homework

- 2, 6d, 7, 12(a,b), 14, 21, 24(d,f,g)
- how to design a even-parity checker?(本章PPT中练习)



Answers

1. d 6. c

2. d 7. b

3. a 8. c

4. c 9. a

5. d 10. c